HSC ENTERPRISE COMPUTING · PEER REVISION

Enterprise School

Six 45-minute lessons on the topics examined most often, each with its own practice set in the exam browser.

Databases · Security · Diagrams · Intelligent systems · Visualisation · 8-mark responses

LESSON MAP

Six lessons, six practice sets

LESSON 1 · SET 1

Relational databases and SQL

- Schemas, PK and FK, 1:many
- Data dictionaries
- SQL SELECT, JOIN, GROUP BY

LESSON 2 · SET 2

Data security and backup

- Matching threats to measures
- Full, incremental, differential
- Backup timelines and restores

LESSON 3 · SET 3

Diagrams

- Context and level 1 DFDs
- System and algorithm flowcharts
- Decision trees, storyboards, UI design

LESSON 4 · SET 4

Expert systems, decision support and hardware

- Structured to unstructured decisions
- Forward and backward chaining
- Choosing hardware for a scenario

LESSON 5 · SET 5

Visualisation, UX, bias and dashboards

- Choosing the right chart
- Data integrity and bias
- Dashboard design

LESSON 6 · SET 6

Big data, ethics and the 8-mark response

- Big data and storage
- Ethical and legal issues
- Planning an extended response

HOW EACH LESSON RUNS

45 minutes, the same four steps every time

10

Teach

The key ideas on two slides

20

Practise

The lesson’s practice set in the exam browser

10

Go over answers

Auto-marks plus the marking guides

5

Exit check

Two questions, answered out loud

Practice sets: chaosontoast.github.io/ec-exam-practice → Practice sets → Enterprise School practice set 1 to 6

Works on phones and laptops. Progress saves in that browser. No account needed.

QUESTION FORMATS

Objective questions

Format	How to approach it	Marks
Multiple choice	Predict before reading the options, cross out two, then pick the best fit.	1
Select all that apply	Judge each option on its own as true or false. Do not stop at the first right one.	1–2, all or most right
Drop-down boxes	Read every row first, do the sure rows, then the rest by elimination.	2: all right, 1: one wrong
Drag to order	Ask what must happen before each step. Usually one card does not belong.	2: all, 1: most in place
Label the diagram	Place the cards you are sure of first. Two cards are usually spare.	2: all, 1: one wrong

QUESTION FORMATS

Tool and written questions

Format	How to approach it	Remember
Spreadsheet	Work out what each cell must show. Use $ on anything that can change. Fill down, then change an input to test.	1 or TRUE() for approximate VLOOKUP, never bare TRUE
SQL	Columns → tables → one join per link → filters → sort. Run it and check the rows against the question.	Tables sit on the left with the stimulus
Short answer	Marks = points. Underline the verb and every “and”: each part needs an answer.	Feature → how → outcome → stimulus
Extended (6–8)	Two-minute plan, one paragraph per part, three stimulus details, end with a judgement.	Quote figures from any table or dashboard

QUESTION FORMATS

Every drawing question and its key rule



Context diagram (level 0 DFD)

One process for the whole system, every entity, labelled flows, no data stores



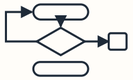
Level 1 DFD

Numbered processes named with verbs; data stores only through a process



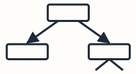
System flowchart

Exam symbols for input, process, storage, document and display



Algorithm flowchart

Start and end terminators, Yes/No on every decision, loops drawn back



Decision tree

Conditions in order, every branch labelled, every path ends in an action



Storyboard

At least four labelled screens in order, each with a short description



UI design (wireframe)

Every field from the stimulus with the right control, annotated with why



Database schema

Every table, PK and FK marked, relationships shown 1 to many

LESSON 1 · IN ALL 6 PAPERS · 6 TO 13 MARKS

Relational databases and SQL

Goal: read a schema, fill in a data dictionary and write a three-table query that runs first time.

Keys link tables, joins go in WHERE

Keys

Primary key identifies each record in its table.

Foreign key holds a primary key from another table, which links them.

One-to-many runs from the PK side (one member) to the FK side (many bookings).

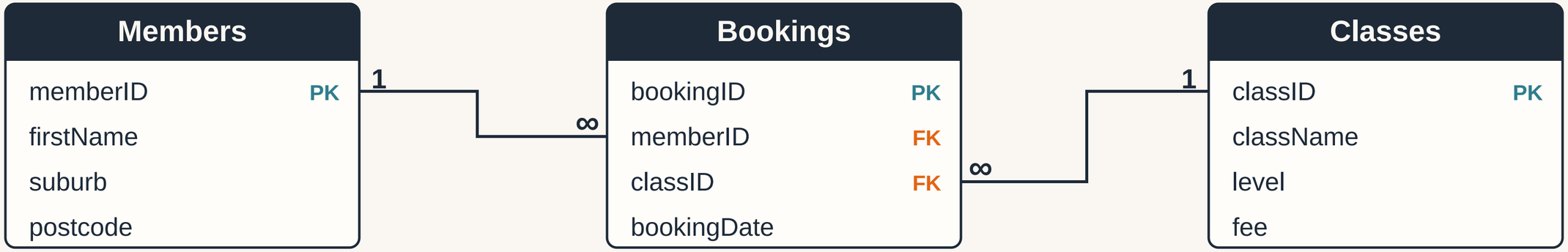
The SQL you need

Only SELECT, FROM, WHERE and ORDER BY are in the course.

One join condition in WHERE **per link**: three tables need two.

```
WHERE M.memberID = B.memberID
AND C.classID = B.classID
```

Reading a schema: one-to-many links



Three tables = two links = two join conditions

```
WHERE Members.memberID = Bookings.memberID AND Classes.classID = Bookings.classID
```

A full SQL answer, laid out like the exam

Members

memberID	firstName	suburb
M01	Ava	Penrith
M02	Ben	Parramatta
M03	Chloe	Penrith
M04	Dev	Ryde

Classes

classID	className	fee
C1	Spin	15
C2	Boxing	25
C3	Yoga	12
C4	HIIT	20

Bookings

bookingID	memberID	classID
1	M01	C2
2	M02	C1
3	M03	C4
4	M01	C3
5	M04	C2
6	M03	C2

List the first name, class name and fee of every booking by a Penrith member, cheapest first, then by first name.

```
1  SELECT Members.firstName, Classes.className,
          Classes.fee
2  FROM Members, Classes, Bookings
3  WHERE Members.memberID = Bookings.memberID
3    AND Classes.classID = Bookings.classID
4    AND Members.suburb = 'Penrith'
5  ORDER BY Classes.fee ASC,
            Members.firstName ASC;
```

1 columns · 2 tables · 3 one join per link · 4 filter · 5 sort

Output after Run

firstName	className	fee
Ava	Yoga	12
Chloe	HIIT	20
Ava	Boxing	25
Chloe	Boxing	25

A data dictionary describes every field

Field	Data type	Format	Size	Why
memberID	Text	X##	3	Letter plus digits
mobile	Text	##########	10	Keeps the leading 0
fee	Currency	$##.##	6	Money
bookingDate	Date/Time	dd/mm/yyyy	10	A date
paidUp	Boolean	Y/N	1	Yes or no only

Field size = count the characters in the format. Digits that are labels (phones, postcodes, IDs) are Text.

BEFORE YOU START · PRACTICE SET 1

How to approach set 1

Questions	Topic	How to approach it
Q1–3	Keys, data types, missing joins	PK identifies a record; FK points to another table's PK. Labels made of digits are Text. Each link needs its own join.
Q4	Data types (drop-downs)	Ask "will I ever calculate with it?" If not, and it can start with 0, it is Text.
Q5	Describe the relationship	Name the type (one-to-many) and the link field: PK in one table, FK in the other.
Q6–7	SQL queries	Columns → tables → joins → filters → sort. Dates are two conditions; put brackets around OR.

PRACTISE · 20 MINUTES

Enterprise School practice set 1

Open

chaosontoast.github.io/ec-exam-practice

Practice sets → Enterprise School practice set 1 → Start

Run each query until its output matches the question.

13 marks

Q1–3 · keys, postcodes, missing joins

Q4 · data types (drop-downs)

Q5 · describe the relationship

Q6 · three-table query with a date range

Q7 · OR with brackets

Mistakes that cost SQL marks

A missing join

Returns every combination of rows, with no error to warn you.

OR without brackets

AND is applied first. Write (a OR b) AND c.

Date ranges

Two conditions: date >= '2026-10-01' AND date <= '2026-10-31'.

Quotes

Text in single quotes, matching the data’s capitals. Numbers without quotes.

EXIT CHECK · OUT LOUD

1. Why is a mobile number stored as text?

2. A query uses three tables but only one join condition. What does it return?

LESSON 2 · IN ALL 6 PAPERS · 3 TO 8 MARKS

Data security and backup

Goal: match a threat to the measure that actually stops it, and justify two measures for a scenario.

Each threat has a measure that targets it

Threat	Measure that targets it
Stolen or guessed password	Multi-factor authentication, strong passwords
Staff seeing data they don’t need	Access levels (least privilege)
Intercepted data, stolen laptop	Encryption in transit and at rest
Phishing, social engineering	Ongoing staff training
Ransomware, hardware failure, disaster	Tested offsite backups, recovery plan
Malware, unauthorised network access	Antivirus, firewalls, updates

Backups: what a restore needs

Full

Copies everything. Slowest to make, simplest to restore.

Incremental

Changes since the **last backup of any kind**. Restore = full + every incremental after it.

Differential

Changes since the **last full backup**. Restore = full + the latest differential.

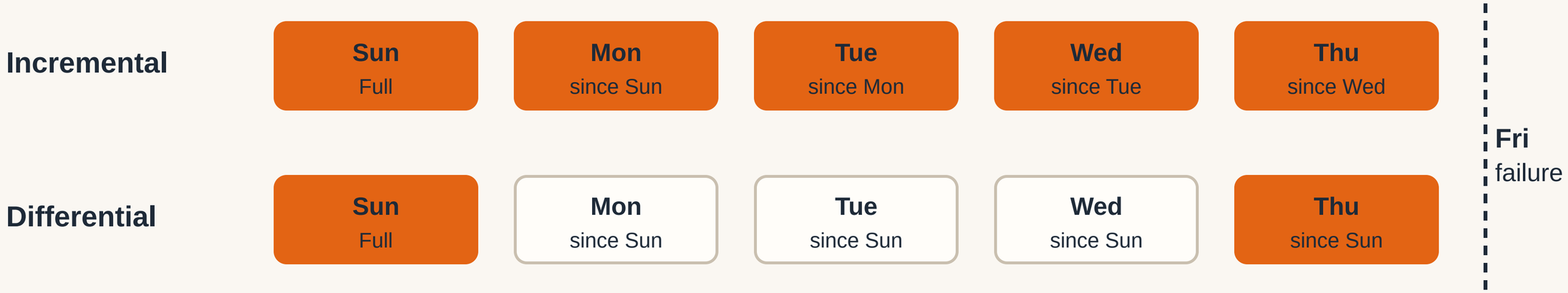
3-2-1 rule

3 copies, on 2 types of media, 1 offsite. Test that restores work.

Risk matrix

Rate each threat by likelihood × impact. Deal with high-high first.

Which backups does a restore need?



needed to restore Thursday night's data

Incremental: full + EVERY incremental after it (smaller, faster backups; slower restore)

Differential: full + ONLY the latest differential (bigger backups each night; faster restore)

BEFORE YOU START · PRACTICE SET 2

How to approach set 2

Questions	Topic	How to approach it
Q1–3	Threats and backups	Match the threat to the measure that stops it. Differential = full + latest; incremental = full + every one after.
Q4	Threat → measure (drop-downs)	Phishing → training, stolen device → encryption, guessed password → MFA.
Q5	Order a restore	Full backup first, then each incremental in date order. Skip anything before the full or after the failure.
Q6–7	Justify measures, risk matrix	Each measure names the threat it stops in the scenario. Rate each threat by likelihood × impact.

PRACTISE · 20 MINUTES

Enterprise School practice set 2

Open

chaosontoast.github.io/ec-exam-practice

Practice sets → Enterprise School practice set 2 → Start

Every measure you write must name the threat it stops.

13 marks

Q1–3 · phishing, compromised accounts, differentials

Q4 · threat → measure (drop-downs)

Q5 · put the restore in order

Q6 · justify two measures (3)

Q7 · use a risk matrix (3)

Mistakes that cost security marks

Listing without linking

“Use encryption and a firewall” earns little. Say which threat in the scenario each one stops and how.

Technology for a people problem

Encryption and firewalls do not stop phishing. Phishing targets people, so training is the direct defence.

Answering half the question

“Implement and investigate” needs both: how it was set up, and how you checked it worked (logs, test restores, audits).

EXIT CHECK · OUT LOUD

1. Sunday full backup, nightly incrementals, failure on Friday morning. Which backups restore Thursday’s data?

2. Which measure limits the damage if one staff account is compromised?

LESSON 3 · IN 4 OF 6 PAPERS · 10+ MARKS

Diagrams

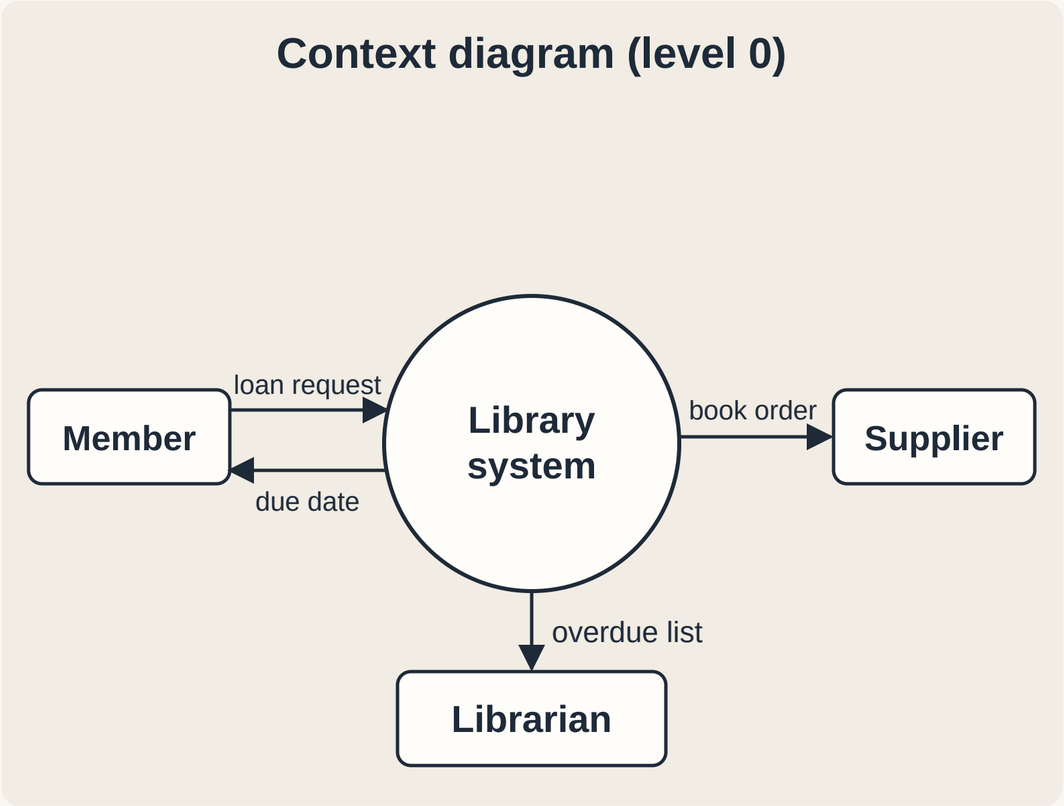
Goal: draw a correct DFD, system flowchart and decision tree from a short scenario.

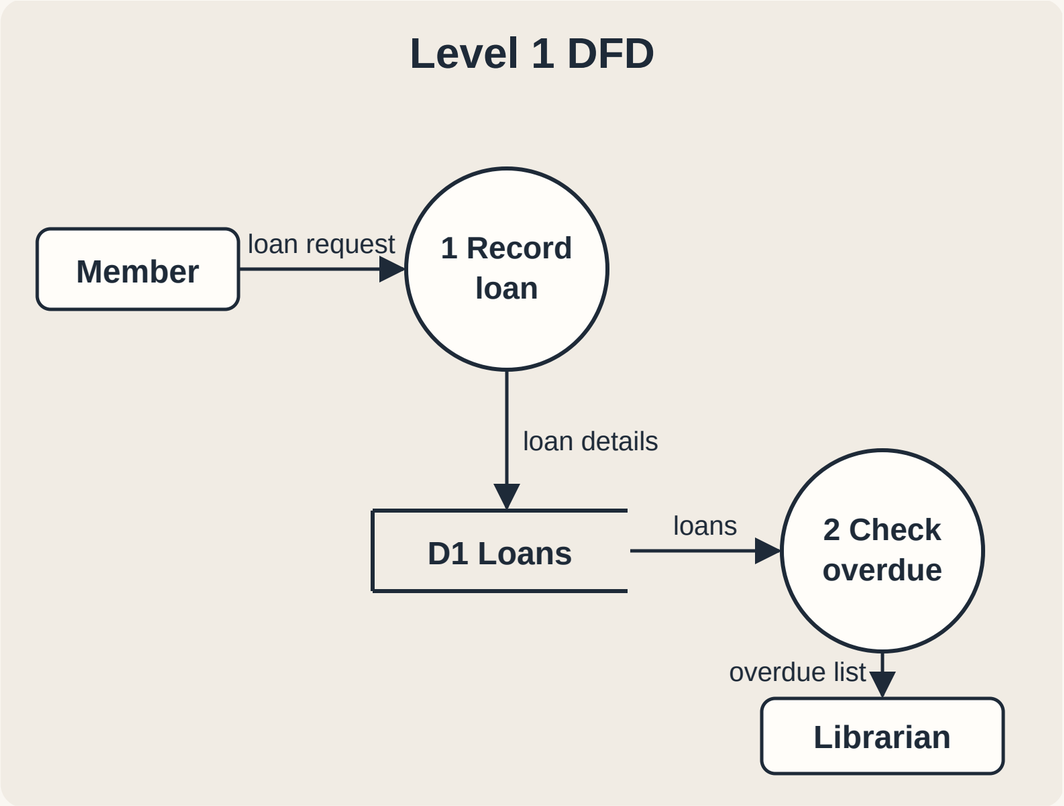
The rules that cost marks when broken

Diagram	Rules
Context (level 0)	One process for the whole system, every entity, every flow labelled, no data stores
Level 1 DFD	Processes named with verbs, data stores reached only through a process
System flowchart	Exam symbols: online input, display, process, document, direct access storage
Algorithm flowchart	Start and end terminators, Yes/No on every decision, loops drawn back
Decision tree	Conditions in order, every branch labelled, every path ends in an action
UI design	Every field from the stimulus, the right control, labelled with why

Context diagram vs level 1 DFD

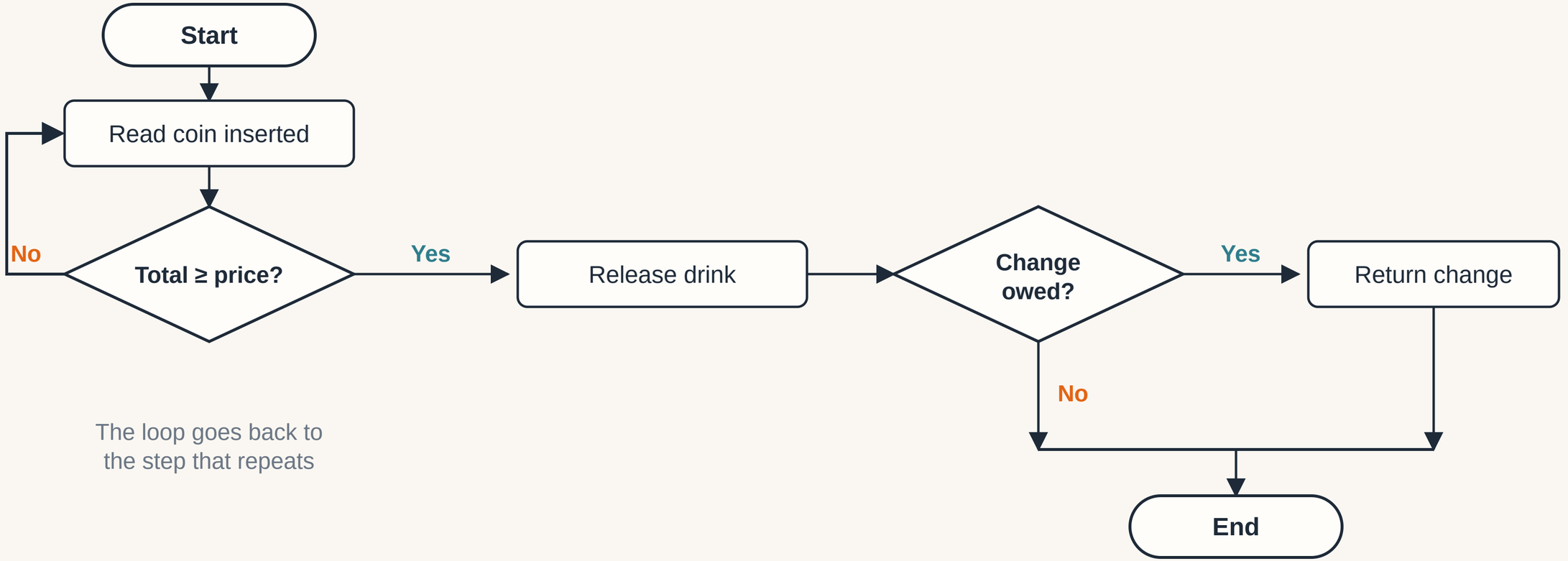
Same library system, two levels of detail





Flowchart rules in one picture

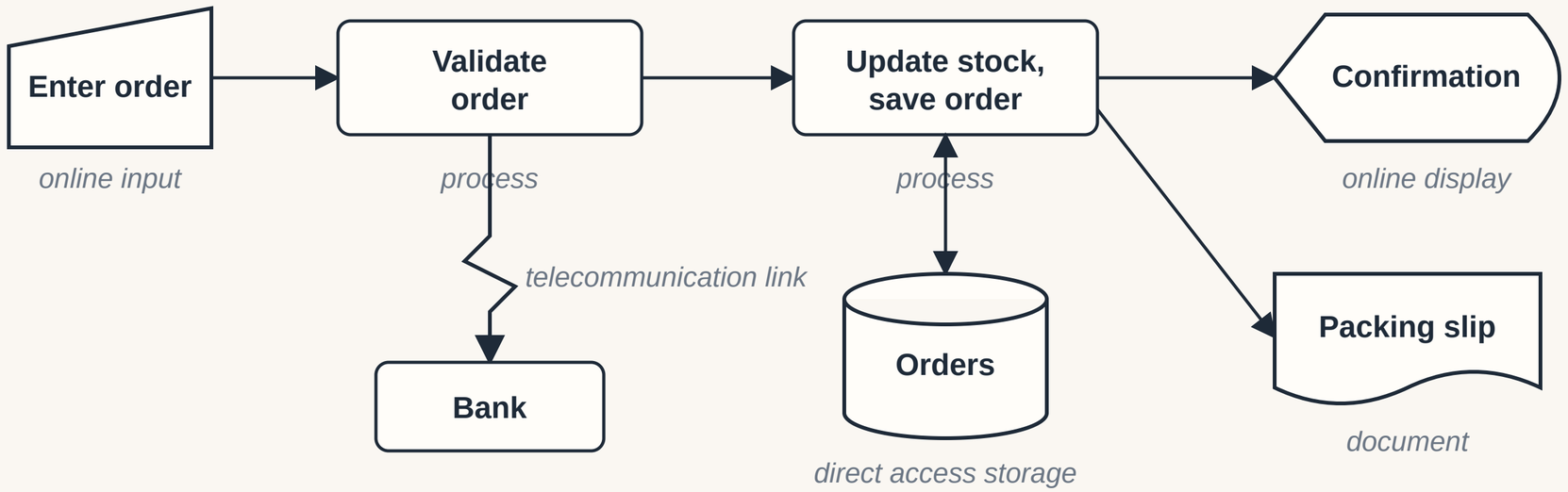
A vending machine: terminators, processes, labelled decisions and a loop



The loop goes back to the step that repeats

System flowchart: an online shop order

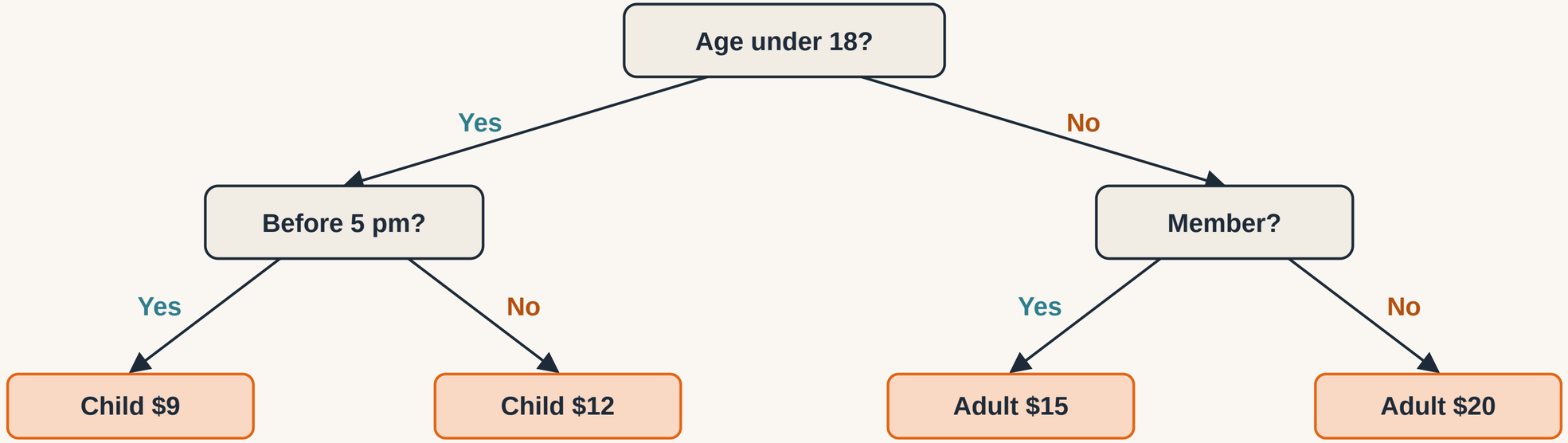
Shows the hardware and files data moves through, not the decision logic



Symbols carry the meaning: name each one in the answer

Decision tree: cinema ticket price

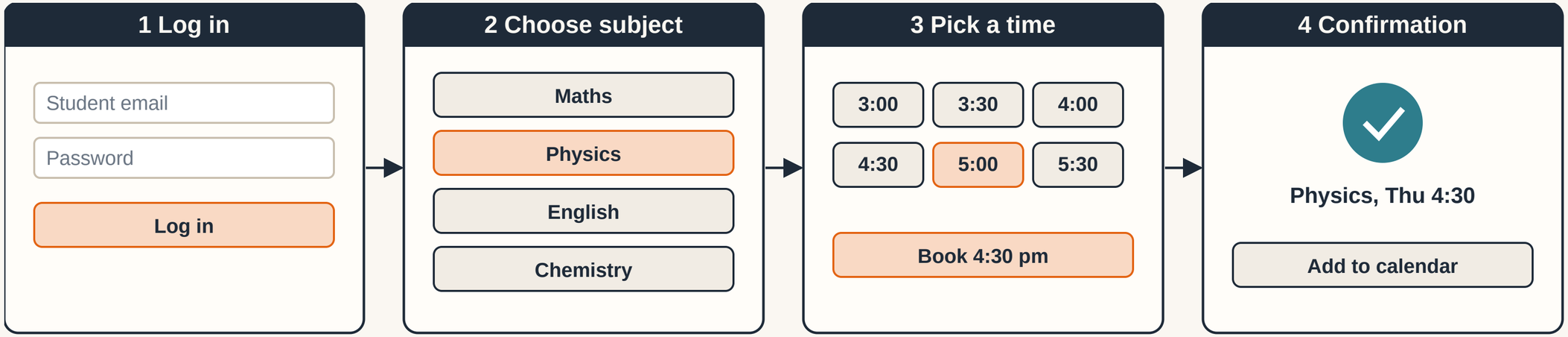
Rule: children $9 before 5 pm, $12 after; adults $15 for members, $20 otherwise



Every branch labelled · conditions in order · every path ends in an action

Storyboard: booking a tutoring session

Four labelled screens, interface elements, a short description and arrows between frames



Student signs in with school email

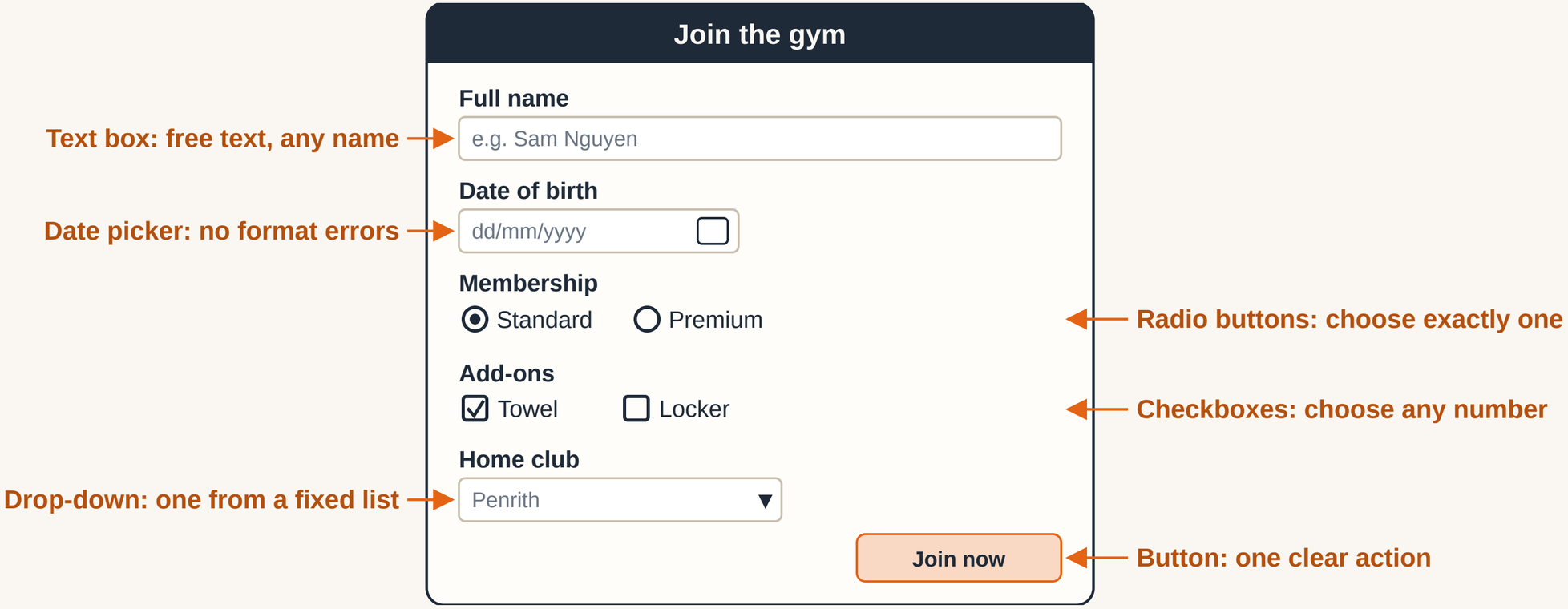
Taps the subject they need help with

Free slots shown; picks one and books

Booking confirmed with a calendar option

UI design: a gym sign-up form

Every field from the scenario, the right control for each, annotated with why



BEFORE YOU START · PRACTICE SET 3

How to approach set 3

Questions	Topic	How to approach it
Q1	Label a decision tree	The first condition goes at the root; every path ends in an action from the scenario.
Q2–3	DFD rules	Context diagram: no data stores. Level 1: stores only through a process.
Q4	Context diagram	One process, every entity from the scenario, every flow labelled with its data.
Q5	Flowchart	“Keeps checking until” = a decision that loops back. Label every Yes and No.

PRACTISE · 20 MINUTES

Enterprise School practice set 3

Open

chaosontoast.github.io/ec-exam-practice

Practice sets → Enterprise School practice set 3 → Start

Then swap screens and mark each other’s diagrams against the rules.

10 marks

Q1 · label the decision tree

Q2–3 · DFD rules

Q4 · context diagram: food delivery app (3)

Q5 · flowchart: car park boom gate (3)

Mistakes that cost diagram marks

Entity to data store

A process must sit between them.

Unlabelled flows

Every arrow and branch carries a label.

No loop back

“Wait until” means a decision that loops to itself.

Missing outcomes

Every outcome in the scenario appears, e.g. save AND send confirmation.

EXIT CHECK · OUT LOUD

1. What is the difference between a context diagram and a level 1 DFD?

2. Which flowchart symbol shows a decision, and what must its branches carry?

LESSON 4 · IN 4 OF 6 PAPERS · INCLUDES AN 8-MARKER

Expert systems, decision support and hardware

Goal: explain how intelligent systems support each category of decision, and name the parts and hardware.

Decision categories are a spectrum

System decides — **Person decides**



Structured

Fixed rules, the system decides

e.g.

Reorder stock below the minimum

Semi-structured

System flags, a person decides

e.g.

Analyst reviews fraud alerts

Unstructured

No procedure, system suggests

e.g.

Board picks a new market

More human judgement as you move right

Three categories of decision

Category	What the system does	Example
Structured	Applies fixed rules and decides	Automatic stock reordering
Semi-structured	Analyses and flags, a person decides	Fraud alerts reviewed by an analyst
Unstructured	Suggests options, no fixed procedure	A board choosing a new market

DSS vs expert system

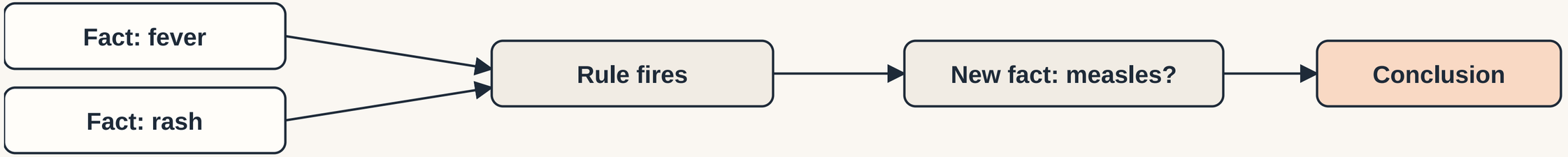
A DSS presents analysis so a person decides. An expert system concludes itself: knowledge base, inference engine, user interface.

Reasoning

Forward chaining starts from facts (monitoring). Backward chaining starts from a goal (testing one suspected cause).

Forward vs backward chaining

Forward chaining (data-driven): start from the facts

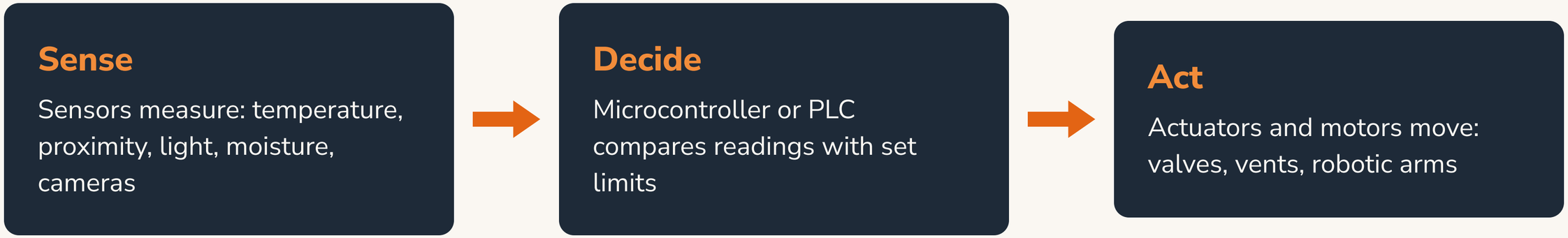


Backward chaining (goal-driven): start from the hypothesis



Suits monitoring and streams of data → forward. Testing one suspected cause → backward.

Hardware: sense, decide, act



The sensor then reports the new reading, so the loop repeats. Other hardware: biometrics, haptics, touch and gesture, voice, VR and AR.

BEFORE YOU START · PRACTICE SET 4

How to approach set 4

Questions	Topic	How to approach it
Q1–3	DSS, reasoning, hardware	DSS: person decides. Goal first = backward chaining. Actuators make things move.
Q4	Decision categories	Ask who decides: system (structured), system then person (semi), person with suggestions (unstructured).
Q5	Order an automated system	Sense → compare with the limit → signal → act → sensor reports back.
Q6–7	Hardware (4), decisions (6)	Name specific sensors. For the 6-marker, one paragraph per category with its stimulus example.

PRACTISE · 25 MINUTES

Enterprise School practice set 4

Open

chaosontoast.github.io/ec-exam-practice

Practice sets → Enterprise School practice set 4 → Start

Follow-up: the 8-mark hospital question at the end of the Peer revision set.

17 marks

Q1–3 · DSS, backward chaining, actuators

Q4 · decision categories (drop-downs)

Q5 · order the greenhouse steps

Q6 · production line hardware (4)

Q7 · airline decisions (6)

Mistakes that cost intelligent systems marks

One paragraph for all three categories

Give each category its own paragraph and its own example from the stimulus.

Categories only for expert systems

The syllabus places the three categories under decision support.

Mixing up inference techniques

“Is a kind of” = ontology. A conclusion withdrawn when a fact changes = truth maintenance.

EXIT CHECK · OUT LOUD

1. Which reasoning method suits a system watching a stream of sensor readings?

2. Which component turns the controller’s signal into movement?

LESSON 5 · IN 4 TO 5 PAPERS · 6 TO 8-MARK DASHBOARDS

Visualisation, UX, bias and dashboards

Goal: assess a dashboard, choose the right chart and explain how a visualisation can mislead.

Pick the chart for the job

Line
Change over time

Bar
Compare categories

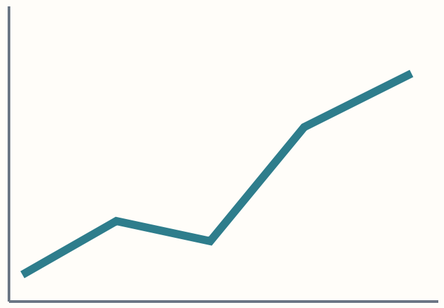
Scatter
Relationship between two numbers

Heat map
Two categories by intensity

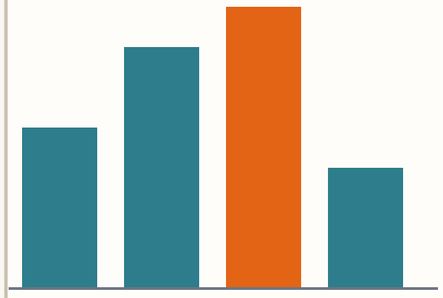
UX judges the user
Time to find a figure, consistency, accessibility (not colour alone), filters, relevance, customisation, live data.

How charts mislead
A truncated axis, missing units, cherry-picked data, colours that are hard to tell apart.

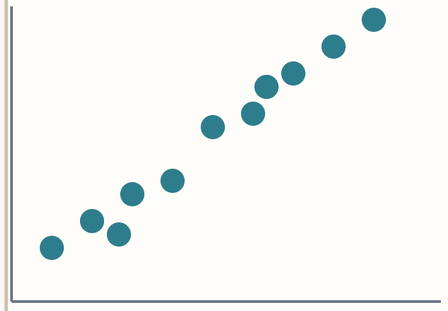
Pick the chart, then check the axis



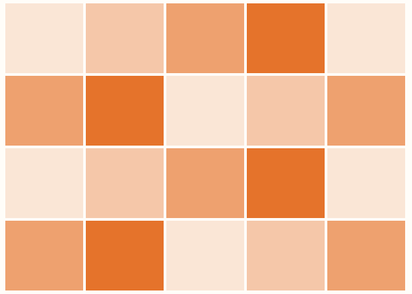
Line: over time



Bar: compare

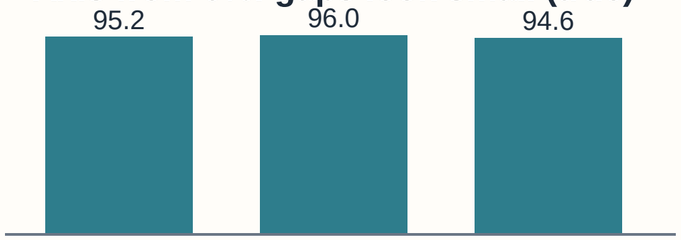


Scatter: relationship

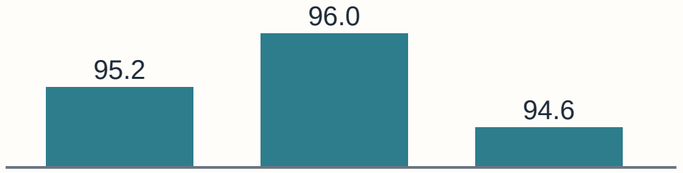


Heat map: intensity

Axis from 0%: gaps look small (true)



Axis from 94%: gaps look huge (misleading)



Data integrity and bias

Ownership

Who is accountable and approves changes

Source

Where the data came from and if it is credible

Validation

Checks that values are sensible

Risk

What could go wrong and its impact

Bias can enter at three points

Collection: who is sampled (use random or stratified sampling, neutral questions)

Storage: what is kept or dropped (validation, no selective deletion)

Analysis: unconscious bias, truncated axes, cherry-picking

BEFORE YOU START · PRACTICE SET 5

How to approach set 5

Questions	Topic	How to approach it
Q1–3	Charts, axes, outliers	Line = time, bar = compare, scatter = relationship. Truncated axes exaggerate. Check outliers before removing.
Q4	Data integrity (drop-downs)	Who is accountable = ownership; where from = source; checks = validation; what could go wrong = risk.
Q5–6	Read and critique a dashboard	Quote real figures. Strengths and limitations must name the chart and its effect on the user.
Q7	PDF tables vs dashboard	Give effects with reasons, and include one downside such as training or clutter.

PRACTISE · 20 MINUTES

Enterprise School practice set 5

Open

chaosontoast.github.io/ec-exam-practice

Practice sets → Enterprise School practice set 5 → Start

Quote a figure from the dashboard in every written answer.

15 marks

Q1–3 · chart choice, axes, outliers

Q4 · data integrity (drop-downs)

Q5 · read the bike share dashboard (2)

Q6 · UX strengths and limitations (4)

Q7 · PDF tables vs a dashboard (4)

Mistakes that cost dashboard marks

Describing instead of using the data

“The line chart shows trips” earns little. “Parramatta rose from 1.2k to 3.4k” earns the mark.

UX words with no user

Say what the user can now do faster or better because of the feature.

Answering half the question

“UX and forecasting” means a full paragraph on each, not one line at the end.

EXIT CHECK · OUT LOUD

1. A chart’s axis starts at 94%. What is the problem?

2. Name the four aspects of data integrity.

LESSON 6 · IN 4 TO 6 PAPERS · 8-MARK ETHICS QUESTIONS

Big data, ethics and the 8-mark response

Goal: plan and write an extended response that reaches the top band.

Big data and the issues it raises

The three Vs

Volume how much · **Velocity** how fast it arrives · **Variety** how many types

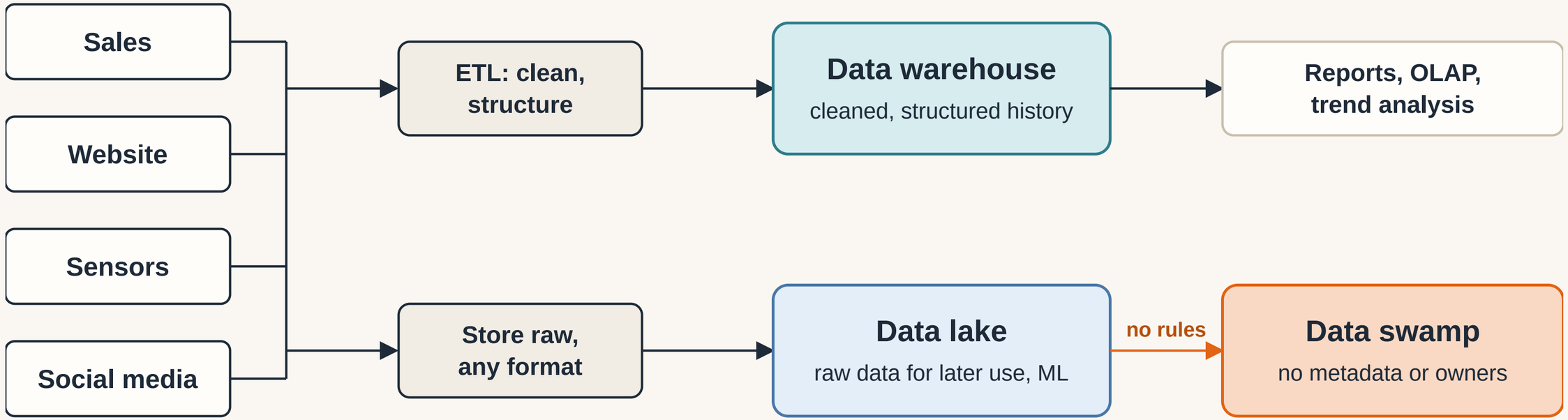
Warehouse cleaned, structured history · **Lake** raw data · **Swamp** a lake with no governance

Social, ethical, legal

Consent and permissions, privacy (Privacy Act 1988, APPs, the OAIC), bias, accuracy, security, copyright, ICIP and Indigenous data sovereignty

An “evaluate” question wants a judgement: weigh benefits against issues and say which wins.

Warehouse, lake or swamp



Velocity, volume and variety are why the lake exists; governance stops it becoming a swamp.

Every point follows the same chain



Two minutes of planning first

Underline every task word. Pick at least three stimulus details. One paragraph per part of the question. End with a judgement.

BEFORE YOU START · PRACTICE SET 6

How to approach set 6

Questions	Topic	How to approach it
Q1–3	Big data terms, the OAIC	Velocity = speed, volume = amount, variety = types. Warehouse = cleaned history, lake = raw.
Q4	Name the issue (drop-downs)	Look for the cue: shared without asking (privacy), unfair results (bias), no credit (copyright), sacred stories (ICIP).
Q5	Evaluate (8 marks)	Plan for two minutes. Several issues, each tied to a stimulus detail and a law or ethical principle, then a judgement.

PRACTISE · 20 MINUTES

Enterprise School practice set 6

Open

chaosontoast.github.io/ec-exam-practice

Practice sets → Enterprise School practice set 6 → Start

Then swap answers and highlight every stimulus detail used. Fewer than three means the answer is capped.

13 marks

Q1–3 · velocity, warehouse vs lake, the OAIC

Q4 · name the issue (drop-downs)

Q5 · evaluate a supermarket loyalty app (8)

Mistakes that cap extended responses

Writing everything you know

Answer the question asked. Content that is not linked to it earns nothing.

Stopping before the outcome

End every point with what the company or user gains or loses.

Only one level

“Individuals or society” questions reward both: one person’s job, then the wider effect.

EXIT CHECK · OUT LOUD

1. What separates the top band from the next one in an 8-mark rubric?

2. Which characteristic of big data is the speed data arrives?

Short on time?

Run lessons 1, 3 and 6. Together they cover databases, diagrams and extended responses, the biggest sources of marks.

Keep practising after the lessons:

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Peer revision set, Peer set 2 and trial papers with built-in marking